DETR-RSI:An Improved RT-DETR-Based Object Detection Framework for High-Resolution Remote Sensing Imagery

Kui Fang
School of Computer Science
South China Normal University
Guangzhou, China
2023023245@m.scnu.edu.cn

Yong Zhan*
School of Computer Science
South China Normal University
Guangzhou, China
*zhanyong@scnu.edu.cn

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***Abstract*—Remote sensing object detection is an important application of computer vision in remote sensing data. However, remote sensing images often exhibit complex backgrounds, densely distributed small objects, and significant variations in object scales, which pose great challenges to detection algorithms. This paper proposes a high-resolution remote sensing image object detection algorithm, DETR-RSI, based on an improved RT-DETR, to enhance the detection accuracy and efficiency of the model on remote sensing images. A lightweight feature extraction backbone network, RPKINet, is designed, a small-object detection layer is added, and a cross-scale feature fusion module, GD-CCFF, is introduced to address the problems of object occlusion and information fusion. Additionally, CIoU loss and Smooth L1 loss are incorporated to optimize the loss function, significantly improving the performance of small object detection. Ablation experiments conducted on the DOTA v1.0 dataset demonstrate the effectiveness of each proposed module. Meanwhile, comparative experiments on the DIOR dataset show that DETR-RSI achieves an average precision (AP) of 57.3%. Compared with various classical object detection algorithms, DETR-RSI also exhibits significant advantages in detection accuracy, especially excelling in the detection of medium and large objects. The source code of DETR-RSI will be made publicly available at https://github.com/Fangkui00/DETR-RSI upon acceptance for reproducibility and further research.**

Keywords-component; Remote sensing, Object detection, RT-DETR

I. INTRODUCTION

Remote sensing object detection is an important application of computer vision in remote sensing imagery. In recent years, with the rapid development of aerospace and UAV technologies, remote sensing images have been widely applied in fields such as mineral exploration, precision agriculture, urban planning, military reconnaissance, and disaster assessment.Compared with natural images, remote sensing images present greater challenges for object detection due to characteristics such as arbitrary object orientations, complex backgrounds, densely distributed small objects, and significant scale variations.

Currently, remote sensing object detection methods mainly include traditional approaches and deep learning-based approaches. Traditional methods often rely on hand-crafted features, generate candidate regions using sliding windows or region proposal mechanisms, and use machine learning classifiers (e.g., SVM or AdaBoost) for object classification and detection. These methods were widely adopted in the early stages of remote sensing object detection.

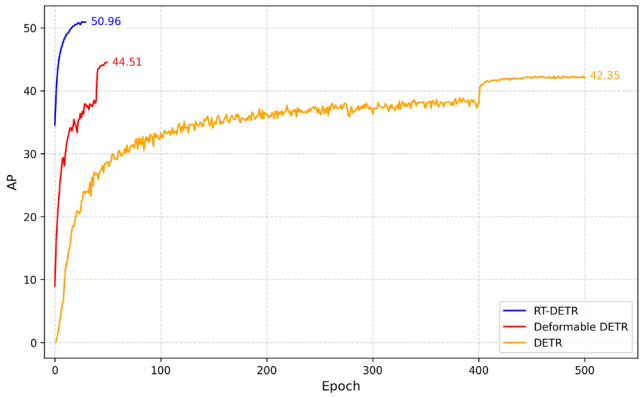


Figure 1. Performance comparison of RT-DETR, DETR, and Deformable DETR on the COCO dataset.

Since the introduction of AlexNet [1], deep learning has rapidly advanced in the field of object detection, gradually replacing traditional methods and becoming the mainstream approach for remote sensing image object detection. Existing deep learning-based detection algorithms are generally categorized into two types: two-stage and one-stage detectors. The former, such as Faster R-CNN [2] and Mask R-CNN [3], generate region proposals followed by classification and regression, while the latter, including the YOLO series [4,5] and SSD [6], directly perform classification and localization on the entire image.

Although these methods have achieved promising results in remote sensing image detection, they still rely on manually designed components such as anchor generation, rule-based target assignment, and non-maximum suppression, which can limit overall performance [7]. To address these issues, Carion proposed DETR, which introduced the Transformer into object detection for the first time. DETR enables end-to-end modeling and simplifies the detection pipeline, attracting widespread attention in the research community.

Recently, Baidu proposed RT-DETR, a real-time object detection model that significantly improves detection speed while maintaining high accuracy [8], effectively alleviating the high computational cost of traditional DETR models. As shown in Figure 1, RT-DETR outperforms both DETR and Deformable DETR in terms of speed and accuracy, demonstrating excellent performance in object detection

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tasks.Given its outstanding performance, this study adopts RT-DETR as the baseline and proposes an improved high-resolution remote sensing object detection method named DETR-RSI. The main contributions of this work are as follows:

To address the challenges of complex backgrounds and large scale variations in remote sensing images, this study designs a lightweight and efficient feature extraction backbone network named RPKINet. RPKINet adopts multiple parallel convolutional branches to effectively suppress background interference, highlight key features, and achieve efficient multi-scale feature extraction.

To tackle the problems of densely distributed and occluded small objects, a dedicated small object detection layer is introduced to enhance scale-awareness and local detail modeling for tiny targets. Additionally, to mitigate the computational overhead and latency introduced by the small-object branch, a cross-scale feature fusion module named GD-CCFF is proposed, enabling more efficient integration of multi-scale information.

In terms of loss function design, the original GIoU loss in RT-DETR is replaced with CIoU, and the L1 loss is replaced with Smooth L1, leading to improved detection accuracy in remote sensing image scenarios.

II. Related Work

Over the past decade, high-resolution remote sensing object detection has received widespread attention, and numerous methods have been proposed to improve detection performance. This section briefly reviews the research progress in this field, with a focus on the development of DETR-based models.

A. Remote Sensing Object Detection

Early remote sensing object detection methods heavily relied on traditional image feature extraction techniques and machine learning classifiers. The Histogram of Oriented Gradients (HOG) features combined with a Support Vector Machine (SVM) classifier have been applied to object detection in remote sensing images [11]. An approach based on Scale-Invariant Feature Transform (SIFT) and a Sparse Representation Classifier (SRC) was proposed for aircraft detection [12]. In addition, Local Binary Pattern (LBP) features have been integrated with deep convolutional neural networks for vehicle detection in remote sensing imagery [13].

In recent years, deep learning-based object detection methods have gradually become the mainstream approach, as they are capable of automatically learning features from data and achieving high detection accuracy in complex backgrounds. Faster R-CNN was further improved by adjusting anchor sizes, fusing multi-level features, and incorporating contextual information, thereby enhancing overall detection accuracy and small object detection performance [14]. YOLO was first applied to high-resolution remote sensing images, reformulating the detection task as a regression problem and greatly improving detection speed [15]. RSI-YOLO, an improved YOLOv5-based model, replaced the original PANet with Bi-FPN to achieve more efficient multi-scale feature fusion [16].

B. DETR-Based Object Detection Algorithms

DETR formulates object detection as a set prediction problem [7], simplifying the training pipeline while achieving promising performance. However, due to its slow convergence and limited capability in detecting small objects, various improvements have been proposed. Deformable DETR introduces deformable attention, which focuses on a small set of sampling points around reference [9] locations to significantly reduce computational complexity. Conditional DETR decouples each decoder query into content and positional components [17], enabling separate modeling of feature and spatial information in the attention mechanism. DAB-DETR [18] introduces dynamic anchors as queries and updates them layer by layer, accelerating training and improving detection accuracy. Lite DETR alternately updates high-level and low-level features, effectively reducing the computational cost of multi-scale feature processing [19].

III. Method

The overall architecture of DETR-RSI is illustrated in Figure 2. As a DETR-like end-to-end detector, DETR-RSI consists of three main components: a feature extraction backbone, a hybrid encoder, and a Transformer decoder enhanced by auxiliary heads. Given an input image, RPKINet is used to extract multi-scale feature maps, which are then fed into the encoder. The hybrid encoder transforms the extracted features into a sequence representation through AIFI and GD-CCFF. A fixed number of features are selected from the sequence as initial object queries. Finally, the Transformer decoder iteratively refines these queries to generate the final detection results.

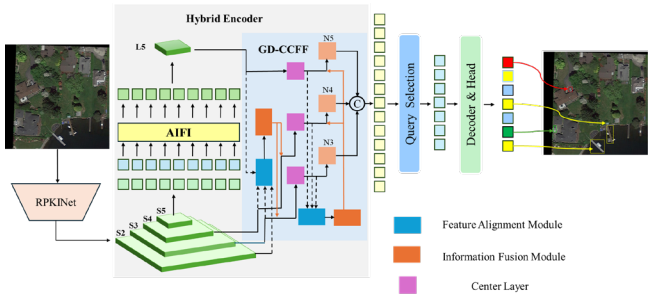


Figure 2. The overall architecture of DETR-RSI.

A. RPKINet

Remote sensing object detection faces challenges such as significant variations in object scales and complex backgrounds, which have drawn increasing research attention in recent years. Based on [10], this paper designs an efficient and lightweight feature extraction backbone network, RPKINet. The overall architecture of RPKINet is inspired by classical backbones such as VGG and ResNet, and consists of a Stem module followed by four stages, as illustrated in Figure 3. Residual connections are incorporated, and each stage adopts a Cross Stage Partial (CSP) structure to enhance feature representation capability. Specifically, the input of each stage is divided into two branches: one is a feedforward network (FFN), and the other consists of multiple PKI modules. The outputs from both

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branches are concatenated to form the final output of the stage, enabling efficient multi-scale feature modeling.

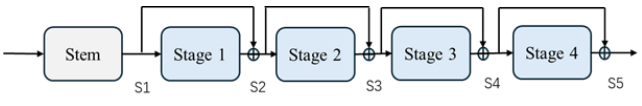
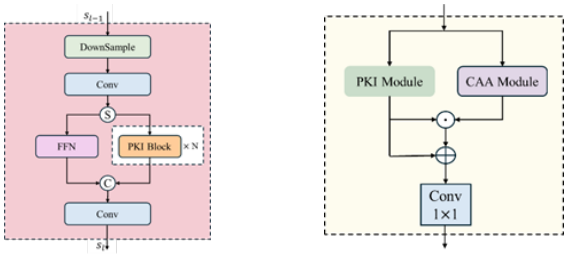


Figure 3. Overall architecture of RPKINet.

The internal structure of a single stage is illustrated in Figure 4(a). The stage input first undergoes a DownSample operation for spatial downsampling and dimensionality reduction, followed by a 1×1 convolution to transform the feature channels. The features are then split into two paths: one passes through a feedforward network (FFN), and the other passes through a series of PKI blocks. The outputs of both paths are concatenated, and the concatenated features are further processed by another 1×1 convolution for feature fusion and channel mapping. The result is the output feature representation of the stage.



(a) Stage l (b) PKI Block

Figure 4. Internal structure of RPKINet.

The PKINet block consists of a PKI module and a CAA module, as illustrated in Figure 4(b). The PKI module is an Inception-style structure, shown in Figure 5. It first applies a 3×3 convolution to extract local information and capture fine-grained details of the target. Then, multiple parallel branches with depthwise separable convolutions are used to extract contextual features at different scales. Finally, a 1×1 convolution is employed to fuse the local features with the multi-scale contextual information, enabling effective integration along the channel dimension. By leveraging convolution kernels of various sizes to extract multi-scale semantics, this structure effectively addresses the significant scale variation of targets in remote sensing images.

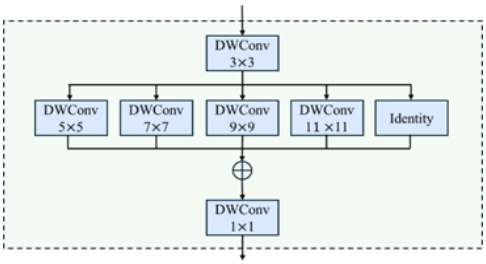


Figure 5. PKI Module

To address the limitations of traditional multi-scale convolution modules in capturing global information, this paper introduces the Context Anchor Attention (CAA) mechanism. The module first applies Global Average Pooling to the input feature map for dimensionality reduction, extracting global contextual features. To further capture long-range dependencies, the CAA module incorporates strip convolutions, which perform depthwise separable convolutions in both horizontal and vertical directions. This design simulates the effect of large receptive field convolutions with reduced computational cost. The introduction of this module effectively suppresses background noise and significantly enhances the representation of key features.[20]

B. GD-CCFF

The original cross-scale feature fusion structure in the RT-DETR model adopts the traditional Path Aggregation Network (PANet), as shown in Figure 6. This structure effectively fuses features from adjacent layers, but features from non-adjacent layers can only be indirectly obtained through step-by-step transmission. As a result, information may be gradually lost during the feature fusion process, which can negatively impact overall detection performance.

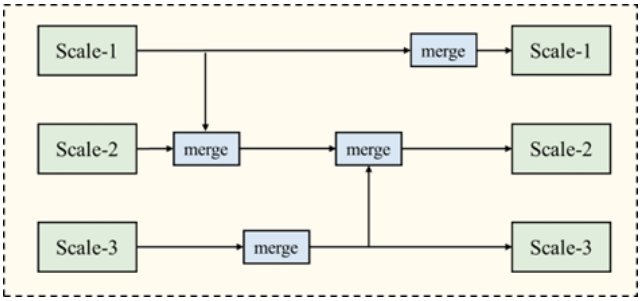


Figure 6. Traditional PANet structure for cross-scale feature fusion.

In remote sensing images, small objects are typically densely distributed and prone to mutual occlusion, posing significant challenges for object detection tasks. Traditional approaches often enhance detection performance by directly introducing additional detection layers for small objects (e.g., the P2 layer). However, such strategies substantially increase the computational burden of the model and impair inference efficiency. Inspired by [21], this paper proposes a novel cross-scale feature fusion module—GD-CCFF—on the basis of incorporating a small-object detection layer. The proposed module effectively integrates fine-grained edge and texture information from shallow features with high-level semantic information from deep features, thereby enhancing information interaction and collaborative representation across different scales. GD-CCFF significantly improves the effectiveness of multi-scale feature fusion and provides strong feature support for the accurate detection of densely distributed small objects.

The GD-CCFF module comprises three key components: the Feature Alignment Module (FAM), the Information Fusion Module (IFM), and the Inject Module. Specifically, FAM is responsible for extracting and aligning features from multiple scales; IFM performs global semantic fusion on the aligned features; and the Inject Module utilizes an attention mechanism to inject the globally fused information back into each feature level. This design not only alleviates the information degradation issue caused by the recursive feature propagation in FPN, but also enhances cross-scale information representation and interaction without introducing significant additional computational overhead.

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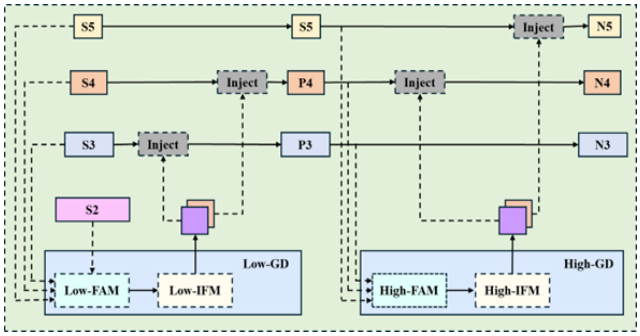


Figure 7. The architecture of the proposed cross-scale feature fusion module

The multi-scale feature maps S2, S3, S4, and S5 extracted by the backbone are fed into the GD-CCFF module, as shown in Figure 7. The Low-FAM module first aligns all feature maps to a common spatial resolution of H/32 × H/32 (matching S4). Higher-resolution features (S2 and S3) are downsampled via average pooling, while lower-resolution features (S5) are upsampled using bilinear interpolation. The aligned features are concatenated along the channel dimension to form a unified representation, denoted as . This is then processed by the Low-IFM module using a re-parameterized convolution (RepBlock) to produce a fused feature map . The fused output is split into two parts along the channel dimension for feature injection.In the Inject module, local and global features are transformed via convolution and integrated through an attention mechanism to enhance representation. Finally, two outputs, P3 and P4, are generated and combined with S5 for the next stage. In this stage, GD-CCFF reuses the same process on P3, P4, and S5 to extract higher-level semantic features and further improve representation quality.

C. Overall Objective

The original RT-DETR model adopts a combination of L1 loss and Generalized IoU loss for bounding box regression, along with focal loss for classification, where the focal loss parameters are set to $\alpha = 0.75$ and $\gamma = 2$. However, the L1 loss suffers from discontinuous gradients and sensitivity to outliers, which can adversely affect training stability. To address these issues, this paper replaces the original L1 loss with Smooth L1 loss, which improves robustness and stabilizes the training process.

In addition, although GIoU enhances the discriminative power of IoU by incorporating the area of the smallest enclosing box of the two predicted and ground-truth boxes, it fails to effectively account for aspect ratio differences. This limitation leads to suboptimal performance when dealing with objects with extreme aspect ratios. To mitigate this, we employ the Complete IoU loss, which simultaneously considers the overlap area, the distance between box centers, and the consistency of aspect ratios. This comprehensive design improves localization accuracy and mitigates over-penalization and instability during training, especially in scenarios with large variations in object shapes and positions. The total loss can be formulated as:

$$L_{total} = \lambda_1 L_{smooth} + \lambda_2 L_{GIoU} + \lambda_3 L_{focal} \quad (1)$$

Following the original RT-DETR configuration, we assign loss weights of $\lambda_1 = 5$, $\lambda_2 = 2$, and $\lambda_3 = 1$ for the Smooth L1, CIoU, and focal losses, respectively.

IV. Experiments

This section first introduces the datasets used, and then conducts ablation and comparative experiments to systematically evaluate the proposed algorithm and verify its performance.

A. Datasets

DOTA v1.0: The dataset contains 2,806 aerial images, most of which have a resolution of 4000 × 4000 pixels, while some range from 800 × 800 to 4000 × 4000 pixels [21]. A total of 188,282 object instances are annotated in the dataset, covering 15 common categories. These objects exhibit significant variations in scale, orientation, and shape, and are distributed in complex and irregular patterns.

DIOR: The images in this dataset are collected from various remote sensing platforms, featuring diverse spatial resolutions, imaging conditions, and geographic regions. DIOR consists of 23,463 optical remote sensing images with 192,472 annotated object instances across 20 common object categories [22]. It is characterized by high inter-class similarity and significant intra-class variability. The objects vary greatly in scale and shape, making DIOR well-suited for the research and evaluation of multi-scale object detection algorithms.

B. Ablation Study

A series of ablation studies were conducted on the DOTA-v1.0 remote sensing dataset to systematically evaluate the effectiveness of each component in the proposed method, as presented in Table I.

TABLE I. Ablation Study Results

RPKI Net	GD-CCFF	New Loss	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L	Params
			53.6	79.0	59.6	36.6	55.9	63.0	40.8M
√			57.2	84.2	64.7	38.9	62.1	64.8	33.0M
√	√		58.8	84.5	66.7	40.2	63.5	66.3	40.2M
√	√	√	59.6	84.9	67.1	41.7	64.0	67.3	40.2M

Starting from the baseline model, introducing the lightweight backbone network RPKINet increased the AP by 3.6 percentage points (from 53.6% to 57.2%) while reducing the parameter count by approximately 7.8M. Further incorporating the P2 layer and the cross-scale feature fusion module GD-CCFF improved the AP to 58.8%, with the AP for medium and large objects (AP_M and AP_L) increasing by 7.6 and 3.3 percentage points, respectively. Finally, adding the improved loss function further boosted the overall AP to 59.6%, with detection performance at all scales improved, including a 5.1 percentage point gain in small object detection (AP_S) compared to the baseline.

C. Comparative Results

To verify the effectiveness of the proposed DETR-RSI model, a series of comparative experiments were conducted on the DIOR dataset. The model was compared with several baseline detectors, including traditional CNN-based methods

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and Transformer-based approaches. The experimental results are presented in Table II.

TABLE II. COMPARATIVE RESULTS ON THE DIOR DATASET.

Method	Epochs	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L	Params
CNN-based models								
RetinaNet	100	29.1	52.9	28.2	2.4	22.5	46.9	35.7M
Faster R-CNN	100	42.3	66.8	45.7	9.0	32.8	62.4	41.5M
Cascade R-CNN	100	45.0	67.0	49.4	9.9	33.9	66.3	53.3M
YOLO v7	300	55.6	76.2	60.5	23.2	46.5	74.4	36.9M
DETR-like models								
DETR	300	46.2	69.1	49.1	8.2	36.0	68.9	41.3M
DAB DETR	100	47.6	71.5	50.8	11.2	37.6	70.6	43.4M
Deformable DETR	100	51.2	75.3	55.3	15.8	42.9	71.3	39.8M
RT-DETR	100	54.7	75.6	59.2	17.3	46.7	72.9	42.7M
DETR-RSI	100	57.3	79.4	61.3	22.2	47.4	75.9	45.7M

In terms of overall performance, the proposed DETR-RSI model achieves an average precision (AP) of 57.3% on the DIOR dataset, outperforming all compared methods. Compared with RT-DETR, Deformable DETR, and the original DETR, DETR-RSI achieves improvements of 3.1%, 6.1%, and 11.1%, respectively. In addition, the model surpasses the powerful CNN-based detector YOLOv7. It is worth noting that although traditional CNN-based detectors such as Faster R-CNN, Cascade R-CNN, and RetinaNet perform stably in natural image detection tasks, their performance is limited on remote sensing images characterized by densely distributed small objects, large scale variations, and complex backgrounds.

In multi-scale object detection tasks, DETR-RSI achieves an APS of 22.2% for small objects, which is slightly lower than that of the CNN-based one-stage detector YOLOv7, but shows a 4.9 percentage point improvement over the baseline RT-DETR, demonstrating the effectiveness of the proposed strategies in enhancing small object detection performance. Compared with small object detection, DETR-RSI performs even better on medium and large objects, achieving 47.4% (AP_M) and 75.9% (AP_L), respectively. This can be attributed to the global modeling capability of Transformer-based detectors, which enables them to capture long-range dependencies between objects and backgrounds.

V. CONCLUSIONS

To address the challenges of dense small objects, large scale variations, and complex backgrounds in remote sensing images, this paper proposes a high-resolution remote sensing object detection algorithm, DETR-RSI, based on an improved RT-DETR framework. The proposed method incorporates a lightweight backbone network (RPKINet), introduces a small-object detection layer and a cross-scale feature fusion module (GD-CCFF), and applies optimized loss functions including CIoU and Smooth L1. These enhancements significantly improve detection accuracy while maintaining computational efficiency.

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